Proceedings of the 7th International Conference on Inventive Computation Technologies (ICICT 2024)
Xplore Part Number : CFP24F70-ART ; ISBN : 979-8-3503-5929-9

Implementation of an NLP-Driven Chatbot and ML Algorithms for Career Counseling

Anuja Deshpande
KIT's College of Engineering (Autonomous)
Kolhapur, Maharashtra, India
deshpandeanuja24@gmail.com

Aryan Dubey
KIT's College of Engineering (Autonomous)
Kolhapur, Maharashtra, India
aryan.udhirdubey2003@gmail.com

Arya Dhavale
KIT's College of Engineering (Autonomous)
Kolhapur, Maharashtra, India
aryadhavale2003@gmail.com

Ankita Navatre
KIT's College of Engineering (Autonomous)
Kolhapur, Maharashtra, India
ankitanavatre25@gmail.com

Uma Gurav
KIT's College of Engineering (Autonomous)
Kolhapur, Maharashtra, India
gurav.uma@kitcoek.in

Amit Kumar Chanchal
KIT's College of Engineering (Autonomous)
Kolhapur, Maharashtra, India
chanchal.amitkumar@kitcoek.in

2024 International Conference on Inventive Computation Technologies (ICICT) | 979-8-3503-5929-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICICT60155.2024.10544511

***Abstract*—The AI-based career guidance system proposed in this study addresses the crucial phase of career decision-making faced by secondary school students in India. Leveraging digital transformation, the proposed system provides an interactive platform accessible to students, aiming to streamline the educational and career selection process. Through extensive aptitude assessments, the system evaluates students' strengths, weaknesses, interests, and aptitudes to generate personalized career recommendations. A 24/7 chatbot support system serves as a virtual career counselor, offering real-time assistance to students. The proposed system's inclusive approach aims to reach students in underserved areas, aligning with the goals of inclusive education outlined in the National Education Policy. Ultimately, the proposed system empowers students to make informed decisions and enable opportunities aligned with their capabilities and aspirations.**

Index Terms*—AI, Chatbot, Career System, NLP, Machine Learning

I. Introduction

Nowadays competition is increasing with the increase in population and technology. Choosing an appropriate career is also important as the world is becoming more technical. Secondary-level students are now more confused about what career they need to choose as there is a vast range of options. After the introduction of the National Education Policy(NEP), it becomes more difficult to choose an appropriate option for their career. It is important to choose a career in which the student has an interest. There is research going on how AI can be integrated in guiding students to choose their careers. Consider a student who has recently completed her 10th grade and finds herself unsure about which career path to pursue. In such a scenario, she can leverage this CareerSprint platform. Here, the student will undertake aptitude, personality, and interest tests, and based on the analysis of these results, a suitable career domain will be suggested to her. Furthermore, she can access a road map for the suggested career through this website, along with the added advantage of learning from this online and free e-resources provided by the system. Additionally, if the student has any inquiries regarding career, he/she can rely on a 24/7 AI-based chatbot for assistance. This system helps students to find out their interests and based on it to choose a career. It starts with the registration of students by providing their name, phone and educational details. Then, students take an interest test and aptitude test and based on a cluster of these, exams results are generated. The system analyses the test results by using Machine Learn- ing algorithms like Decision trees and clustering techniques and provides reports of both tests separately. Then system will do a combined study of both test results and on this, it will recommend a suitable career for the student along with a detailed career path. It also has a 24/7 chatbot for interaction with the student to ask queries in their mind about recommended careers. The main core of the system is educational resources, this feature of our system provides students educational resources to achieve their goals along with some educational blogs. In this respect, students can explore the undiscoverable paths that suit their strengths and aspirations. The system acknowledges the need for practical guidance and links the students to professionals working in their fields of interest. It is instilled in students that they can always grow and they have the means to develop themselves all their lives.

II. Related Works

Many of the existing systems use AI and ML to mainly cover or recommend careers for graduates or recommend jobs for them but there is a lack of research for secondary-level students.

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.

Traditional one-to-one counseling gives better results but is not feasible for a large number of students. Nowadays students are more confused about which career they should choose.

In [1] the Career Guidance System employs a multifaceted approach, combining registration and profiling, tailored aptitude testing, expert advice, and AI-driven techniques such as NLP and machine learning. It offers personalized guidance, accessibility, and data-driven decision-making while relying on expert insights. However, limitations include data quality dependency, a lack of real-world experience, limited scope, potential cultural bias, and the risk of over-reliance on technology. [2] presents a mentoring chatbot using Flowxo and Dialogflow, offering students 24/7 accessibility, privacy, and efficiency. While it lacks human empathy and may struggle with complex scenarios, the study shows generally satisfactory user experiences, especially for academic-related inquiries, highlighting the potential for further enhancement and development. [3] introduces a Smart Career Guidance and Recommendation System, leveraging machine learning techniques to provide personalized guidance and skill certification suggestions. While offering data-driven decision-making and improved student-advisor relationships, the system faces limitations concerning data collection, algorithm selection, data quality, and its narrow focus on student attributes. Continuous refinement and a broader perspective on career factors are critical for its continued enhancement. [4] utilizes machine learning and psychometric tests for automated career guidance. It operates through a client-server model with an Android app and Node.js server, offering user-friendly interaction and accurate predictions. However, data quality, limited career streams, privacy concerns, operational costs, and the need for continuous improvement are vital considerations for its effectiveness and user satisfaction. [5] presents an intelligent career guidance system using the K-Nearest Neighbor algorithm and a wide array of student parameters, offering personalized recommendations. Despite the promise of data- driven advice, accuracy concerns, limited parameter inclusion, and the lack of real-time capabilities pose challenges. The system's focus on traditional students and the necessity to address outliers warrant ongoing refinement for broader applicability. [6] offers personalized advice and a user-friendly interface for students. While its 24/7 availability and potential for expansion are strengths, it faces challenges related to data quality, accuracy, scalability, and dependence on NLP. Ongoing updates and careful dataset creation are essential for its effectiveness in guiding students' career decisions. [7] introduces an AI-driven career guidance system with a method involving data collection, analysis, and the use of an Artificial Neural Network (ANN). It offers efficient and personalized recommendations while reducing human bias, but limitations include data quality, historical data reliance, lack of human interaction, and technological access constraints. Adaptation to evolving career trends is crucial for its effectiveness. [8] presents a career counseling framework leveraging ML and AI techniques for data-driven, personalized guidance. While offering interpretability and performance comparison benefits, limitations include a limited dataset, data quality considerations, and the potential for further exploration of socio-economic factors and unsupervised learning techniques. Ongoing research and data enhancement are essential for the system's continued development. [9] presents an Intelligent Career Guidance System utilizing machine learning to enhance career recommendations. While offering data-driven and accurate guidance, limitations include data quality and the need for system improvements. Future iterations should focus on expanding dataset quality and considering additional influencing factors for comprehensive career counseling. [10] underscores the potential of the KNN algorithm in providing personalized career guidance for students based on their interests and past courses. While highlighting its simplicity and data-driven approach, it acknowledges challenges related to dataset size, the choice of the 'k' value, and distance metric considerations. The study emphasizes the need for further optimization to enhance the algorithm's effectiveness in education and career counseling [11] is an AI-based chatbot system for career counseling that offers user-friendly and accessible support in resolving career-related queries. While it holds promise in pro- viding guidance, potential limitations may include data quality, scope constraints, technical issues, and the absence of human interaction, which should be considered in its implementation. Further details about these limitations are not extensively covered in the text. [12] explores the application of machine learning in predicting students' academic performance, enabling early intervention and personalized education. While this approach holds potential for optimizing educational re- sources and decision-making, it must contend with challenges related to data quality, interpretability, ethics, and system complexity. Additionally, the accuracy of predictions relies on high-quality, non-noisy data from student surveys. [13] The paper introduces a conceptual model for a hybrid neural expert system aimed at improving career counseling for tenth-grade students in Oman. The system integrates expert systems and neural networks, incorporating career assessment instruments to predict suitable career paths based on personality traits. The methodology involves knowledge acquisition, validation, modeling, and the creation of a knowledge-based system with an inference engine, while the integration of neural networks enhances the predictive capabilities. The proposed hybrid system seeks to overcome limitations, but potential advantages and limitations are not explicitly outlined. [14] The research paper describes the development of an AI chatbot to support admissions and career guidance for universities. The methodology involves building a structured dataset using natural language processing and machine learning to create identification models. Tools such as HTML analysis, text mining, and SVM models are applied. The advantage lies in providing 24/7 automated support, improving the quality of advice, and overcoming repetitive scenarios in admissions counseling. However, limitations include the need for continuous enrichment of the dataset for higher accuracy.

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.

[15] The creator discusses a task suggestion framework. Here AI (Cooperative sifting & Vector space model) and data mining methods were applied to a Soothing web server application that overcomes any barrier between the frontend (android application) and the backend (MongoDB case) utilizing APIs. The information imparted through APIs is taken care of in the data set and the Proposal Framework utilizes that information to blend the outcomes. [16] The creator utilizes the information of existing understudies, where the characters, inclination, and general data of the understudy are planned with their professions. The Instinctive Vocation Framework utilizes different inquiries that understudies need to pay all due respects to test their fitness as well as inquiries on the understudy's experience. The responses are then placed into the made model to foresee the vocation that matches the understudy's inclination and character [17] research explores the role of chatbots in supporting students' careers, personalized information counselor augmentation, and direct counseling. [18] Presented a research paper that introduces a web-based counseling system with a chatbot helping students to gain information, competitive exam details, quizzes, and a counseling chatbot with user-friendly interfaces. [19] [20] [21] presented the role of machine learning in Natural language processing using investor's sentiments to do stock market analysis. [22] [23] presented analysis and Comparison of DC-DC converter with the help of soft Computing machine learning algorithms. A summarization of related work is presented in the Table I.

TABLE I: LITERATURE REVIEW

Sr. No.	Authors	Applications	Methodology	Observations
1	Divya Manoj Bhushali, Prashant Itankar	Career counselor system with integration of chatbot	Registration, profiling, aptitude testing, expert advice, NLP, ML	The project is an overreliance on technology; chatbot recommends career on test not able to answer all queries.
2	Sanchi Satam, Tejal Nimje and co.	Chatbot for career guidance	Flowxo, Dialogflow	It helps students to overcome anxiety, stress but it struggles in complex scenarios
3	Lakshmi Prasanna, Dr. D. Haritha	Smart career guidance and recommendation system	Machine learning techniques	Data collection, algorithm selection, data quality, narrow focus on student attribute
4	Shivam Mathwad, Rushikesh Pede and co.	Career recommendation system	Machine learning, psychometric tests,Android app,Node.js server	Data collection, operational cost is high
5	Sneha H S, Shreya R and co.	Machine learning based career guidance system	K-Nearest Neighbor algorithm	Accuracy concerns, limited parameter inclusion, lack of realtime capabilities
6	Ritik Raut, Ritesh Sharma,Riteek Kharbade	AI chatbot for career counseling	NLP	Data quality, accuracy, scalability, dependence on NLP
7	Firdosh Sayyed, Ronak Sanghani and co.	Career guidance	Data collection, analysis, Artificial Neural Network	Data quality, historical data reliance, lack of human interaction, technological access constraints
8	Guleria and Manu Sood	AI and ML-based career counseling	ML and AI techniques	Limited dataset, data quality, further exploration of socio-economic factors, unsupervised learning techniques
9	Vignesh S, Shivani Priyanka C and co.	ML-based career counseling	Machine learning techniques	Data quality, system improvements
10	Indriyani, Laros Tuhuterub and co.	KNN algorithm precisely give career recommendations	KNN algorithm	Additional features can be added
11	Danyal Zaidi Sahil Raza Lalit Sharma	Career counseling chatbot	AI-based chatbot system	Data quality, scope constraints, technical issues, absence of human interaction
12	Radhika R Halde	Education system betterment	Machine learning al7gorithms	Data quality, interpretability, ethics, system complexity
13	Khoula Al-Abri, Manjit Sidhu	Neural network-based career counseling	Hybrid neural expert system	Model is not correctly working, and data is insufficient
14	Le Hoanh Su, Truong Dang-Huy and co.	Chatbot for university admission and career counseling	AI chatbot development	Continuous enrichment of the dataset for higher accuracy
15	Harsh Jain, Misha Kakkar	Job recommendation	Machine learning, data mining, RESTful API, Android IDE	Used a different approach for job recommendation
16	Rucha Rangnekar, Khyati Suratwala and co	Career prediction	Data mining, linear classification	Derived personality from interest
17	Terri Lee,Krithika Jagannath,Nitin Aggarwal	Chatbot	Interactive chatbot interviews	Used chatbot for interviews and suggested career path
18	Ganesh Dongre, Farooque Vadraiwala and co.	Self assessment through quizzes	Web based chatbot	Provides web base career counseling system with quizzes and chatbot.

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.

- **Limitations and Issues of existing models**

1) Some models rely only on chatbots and predict careers for students by taking quizzes and interviews through chatbots.
2) Some models use machine learning but they lack data quality and are unable to predict the ultimate career path.

- **Research Contribution**

The novelty of this project is that no prior work has considered all parameters.

This system mainly focuses on:

1) Educational resources and roadmaps in a particular domain.
2) Aptitude test, Interest Test, and personality test for students.
3) Instead of a traditional counselor this system has virtual interactive AI chatbot that acts as a counselor.

III. METHODOLOGY

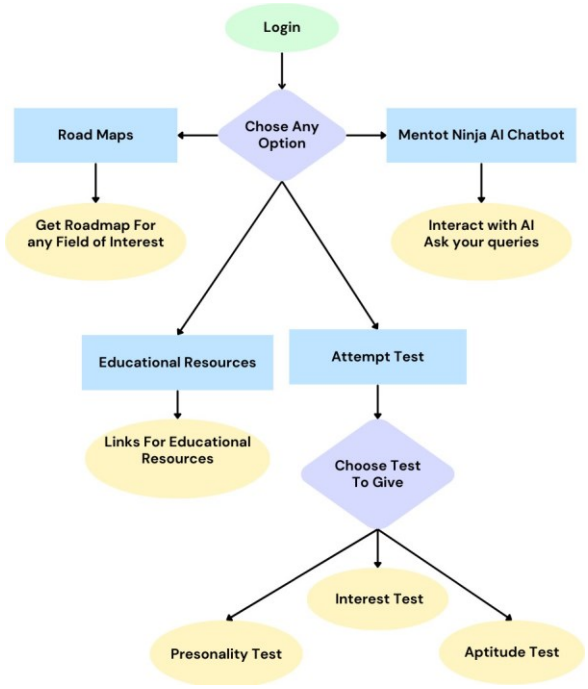


Fig. 1: Block Diagram

The main objective of the proposed system is to guide students in village areas about their careers and give them detailed career paths. The proposed system has many modules which will be explained later. It integrates an intelligent chatbot for solving queries of students and also aligns with NEP. The system has user-friendly interface.

In the first module of this web application, students can securely register and log in using HTML, CSS, JavaScript, and Flask server for MySQL database integration. This module ensures the confidentiality of students' information, providing a foundation for subsequent features. Following registration, the platform initiates data collection and analysis, encompassing academic records, aptitude test results, and interest assessments. This comprehensive approach enables us to gain insights into students' strengths, interests, and preferences, forming the basis for personalized career guidance.

The core functionality of the AI-driven platform lies in the Interest Assessment and Aptitude Testing modules. Through an engaging interest test, students express their passions, receiving intuitive graphical insights into potential career paths aligned with their interests. The aptitude test provides a holistic assessment, visually presenting students' strengths and areas for improvement. AI-driven 24x7 chatbot is also present to help out students with their doubts. They can ask them queries, and thoughts as it serves as a counselor in the system. Along with it this system also consists of roadmaps for education after secondary school and educational resources. for them.

To present the recommended careers and learning paths, the system platform employs a machine-learning model known as a Decision Tree. The output from this Decision Tree model is visualized through a graph, providing a clear and comprehensive representation of the suggested career paths based on the individual's aptitude, personality, and interest test results. This graphical approach enhances the user experience by offering a visual roadmap for their career journey.

A. Web Portal

The diagrams Fig 2, Fig 3, Fig 4, Fig 5 below show how the web application looks:

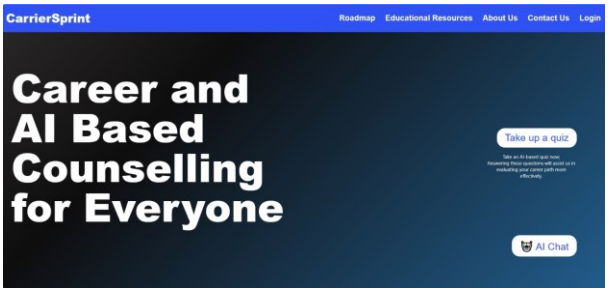


Fig. 2: Home Page

Fig 2 shows the home page of the system. Which consist of different buttons like Roadmap, Educational Resource, about us, contact us, Take a quiz, AI Chat etc.



Fig. 3: Roadmap Page

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.

Fig 3 shows the roadmaps to the student about various careers.

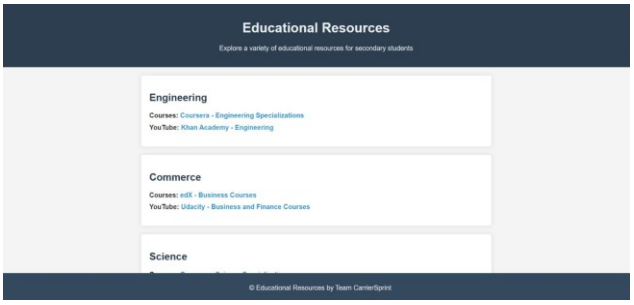


Fig. 4: Educational Resources Page

Fig 4 shows the educational resources for students through which they can get a free online lecture for entrance exams, Regular semester exams, etc.

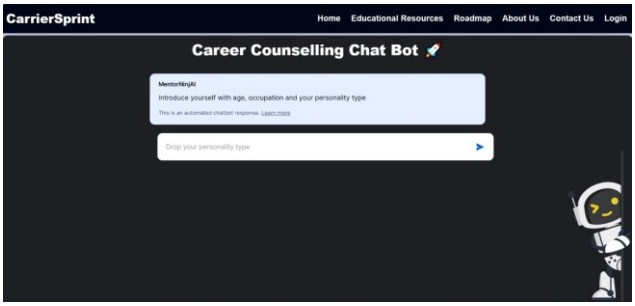


Fig. 5: Chatbot Page

Fig 5 shows a chatbot through which students can interact and ask their doubts.

B. Machine Learning

Machine Learning (ML) is used to predict and recommend suitable careers for students. Various machine learning algorithms including Decision Tree, Adaboost Classifier, KNN, and Naive Bayes Classifier are considered.

- Decision Tree

A decision tree is the supervised learning algorithm used for both categorical and continuous variables. In this system, researchers have used a decision tree to find out which label the data belongs to. Nodes of the decision tree denote the input variable X and leaf nodes denote the output variable(Y). The decision tree calculates information gain and entropy for each node. Entropy is the measure of randomness of data and information gain checks how much entropy is reduced. It is represented as:

$$Entropy(S) = -\sum_{i=1}^{c} p_i \log_2(p_i) \quad (1)$$

$$IG(S,A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) \quad (2)$$

A decision tree classification is all about finding an attribute that returns the highest information gain. If the information gain is 0, it means that no reduction in entropy due to the split. When the information gain is maximum, entropy is reduced. Values are calculated using this entropy and information gain. These values are responsible for achieving the classification process.

- Adaboost Classifier

Adaboost Classifier is an ensemble technique used for both classification and regression analysis. It is used to classify multiple weak or base learners into a stronger learner.

- KNN classifier

KNN is a supervised learning classification algorithm that is mainly used for multiclass classification. In classifications output has labels. KNN is the simplest algorithm in machine learning. KNN is used here to suggest to students from their given input which career they have to opt for. The equation to get a suitable career choice for a student is :

$$Distance = \sqrt{(x1 - x2)^2 + (y1 - y2)^2} \quad (3)$$

- Naive Bayes Classifier

Naive Bayes is a supervised learning algorithm used to classify data based on probability. It is a classification algorithm based on Bayes theorem. In this classifier, it is assumed that features are conditionally independent of a given class label. These features include interest, value, skill, and personality for career class. The probability equation of Bayes theorem is:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (4)$$

From the above, all algorithms used for the prediction Decision tree is the most accurate model with an accuracy of 0.95. Hence decision tree is used as the model for the system.

C. Role of NLP for Chatbot

The chatbot in the proposed system acts as a counselor for the student's queries. The chatbot is built on NLP which enables the chatbot to understand and interpret student inputs. When the student asks some Query to the chatbot, then that query will be sent to the NLP Processor for processing. Then NLP Processor compares the query with queries already stored in a database. If the queries have not found any match with the queries stored in a database. Then that query will send for extraction by which it is extracted in some entities an text or keywords. Then the extracted keywords or entities are processed to find a pattern.

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.

IV. Result and Analysis

The proposed system includes the use of various machine learning algorithms and an AI chatbot that is capable of giving answers to any query coming from a student.

The performance of classification models is determined using a confusion matrix and accuracy plot. A confusion matrix is used to evaluate the performance of a classification model in machine learning representing their accuracy. It gives the number of True Positives(TP), True Negatives(TN), False Positives(FP), and False Negatives(FN).

1) True positives (TP) are the values in which the examples are correctly determined as positive
2) False positives (FP) are the values in which the examples are negative but are determined as positive.
3) True negatives (TN) are the values in which the examples are correctly determined as negative
4) False negatives (FN) are the values in which the examples are positive but are determined as negative.

Fig 6 shows the confusion matrix for the proposed model.

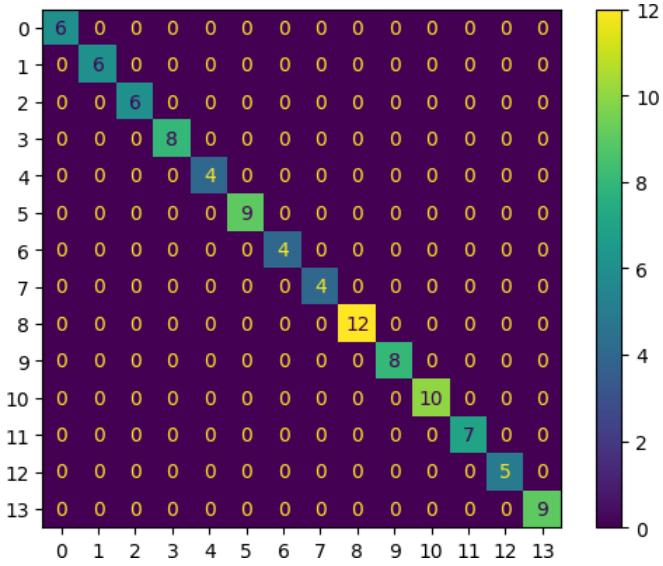


Fig. 6: Confusion Matrix

Fig 6 shows the confusion matrix of the model which gives all the predictions true because of the limited dataset.

A confusion matrix is used to determine the performance measure of the model by evaluating accuracy, precision, recall, and f-measure. The following are the formulas for all the above-mentioned measures:

Precision = TP/(TP+FP)

Recall = TP/(TP+FN)

Accuracy = (TP+TN)/(TP+TN+FP+FN)

F-measure=2 * (precision*recall)/(precision+recall)

Error- rate = 1-accuracy

Following TABLE IV shows performance measures for classification techniques:

TABLE II: The performance measure for classification models

	Accuracy	**F-measure**	**Error Rate**
Decision Tree	1.0	1.0	0.00
KNN	0.91	0.908	0.09
Adaboost Classifier	0.29	0.286	0.71
Naive Bayes	0.93	0.93	0.07

From the above TABLE IV Decision Tree is the classification algorithm best suited for recommending a career.

The diagram represented in Fig 7 is the accuracy plot of the algorithms used to predict suitable career choices for students. The accuracy of all algorithms used for the work is plotted using accuracy(), the standard metric from the sklearn library.

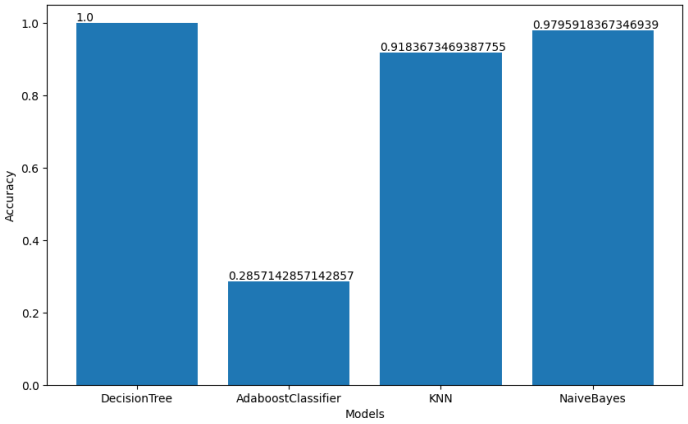


Fig. 7: Accuracy Plot

Fig 7 shows the accuracy plot of comparative analysis of all algorithms used in the system. It shows among all algorithms decision tree has the highest accuracy of 1.0. This accuracy is achieved due to a limited dataset.

High accuracy can be achieved by increasing training data with precise and clean relevant data. Researchers can embrace cross-validation to evaluate and increase models' performance and incorporate error analysis to identify areas for improvement.

There is scope for adding more features to the dataset. In the future researchers can add multilingual support for a chatbot which can be beneficial for ground-level students as of now we can ask queries only in English. Along with it gamification and rewards also be added. The career guidance system can be aligned with the vision of inclusive education outlined in the National Education Policy in the future.

V. Conclusion

In developing the system, researchers have crafted a web-based application aimed at guiding high school students in making informed career choices. The system prioritizes user-friendliness and interactivity, employing digital technology to revolutionize the educational and career decision-making

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.

journey. A key feature is the integration of an AI chatbot, acting as a virtual counselor to provide real-time, personalized responses. It suggests the best careers based on their interests and skills. The system utilizes a decision tree classifier, conducting tests to recommend the best career path for students based on their interests and aptitudes. Notably, the researcher's observations reveal varying accuracy scores among different algorithms, with the decision tree classifier showing the highest accuracy. Looking ahead, the system's potential for growth is exciting. Future enhancements may involve the addition of new features, aiming to further enrich the knowledge and experience of students as they navigate their educational and career paths.

REFERENCES

[1] Prashant Itankar Divya Manoj Bhnushali. Online career counselor system based on artificial intelligence: An approach. International Organization of Research Development (IORD, 2020.

[2] Sanchi Satam, Tejal Nimje, Shreya Shetty, and Samidha Kurle. Mentoring chatbot using artificial intelligence framework. S-JPSET, 2020.

[3] Lakshmi Prasanna and Dr. D. Haritha. Smart career guidance and recommendation system. IJEDR, 2019.

[4] Shivam Mathwad, Rushikesh Pede, Aniket Chaudhari, Prashant Patil, and Mr. Nikhil Dhavase. Career recommendation system. International Journal of Engineering Research in Computer Science and Engineering (IJERCSE), 2021.

[5] Sneha H S, Shreya R, Sahana B C, Soumya Sangalad, and Dr. Paramesha K. A survey on intelligent career guidance system using machine learning. International Research Journal of Modernization in Engineering Technology and Science, 2022.

[6] Ritik Raut, Ritesh Sharma, and Riteek Kharbade. Krishna – the career guidance chatbot. International Research Journal of Engineering and Technology (IRJET), 2021.

[7] Firdosh Sayyed, Ronak Sanghani, Abhishek Vora, and Nikita Lemos. Ai-based career guidance. International Research Journal of Engineering and Technology (IRJET), 2020.

[8] Guleria and Manu Sood. Explainable AI and machine learning: performance evaluation and explainability of classifiers on educational data mining inspired career counseling. Springer, 2022.

[9] Vignesh S, Shivani Priyanka C, Shree Manju H, and Mythili K. An intelligent career guidance system using machine learning. IEEE, 2021.

[10] Indriyania, Laros Tuhuterub, Gentur Wahyu Nyipto Wibowoc, Alex Wendad, and Nengah Sandie. A study on the effectiveness of the k-nn algorithm for career guidance in education. JINAV: Journal of Information and Visualization, 2022.

[11] Danyal Zaidi, Sahil Raza, and Lalit Sharma. Artificial intelligence-based career counselling chatbot a system for counseling. Annals of R.S.C.B, 2021.

[12] Radhika R Halde. Application of machine learning algorithms for betterment in the education system. IEEE, 2016.

[13] Khoula Al-Abri1 and Manjit Sidhu. Conceptual model of hybrid neural expert system for career counselling. International Journal of Application on Sciences, Technology and Engineering (IJASTE), 2023.

[14] Le Hoanh Su1, Truong Dang-Huy2, Tran Thi-Yen-Linh2, Nguyen Thi-Duyen-Ngoc2, Ly Bao-Tuyen2, and Nguyen Ha- Phuong-Truc3. Development of an ai chatbot to support admissions and career guidance for universities. International Journal of Emerging Multidisciplinary Fluid Sciences, 2020.

[15] Harsh Jain and Misha Kakkar. Job recommendation system based on machine learning and data mining techniques using restful API and android ide. IEEE 9th International Conference on Cloud Computing, Data Science Engineering, 2019.

[16] Rucha Rangnekar, Khyati Suratwala, Sanjana Krishna, and Dr. Sudhir Dhage. Career prediction model using data mining and linear classification. IEEE Fourth International Conference on Computing Communication Control and Automation (ICCUBEA), 2018.

[17] Nitin Aggarwal Terri Lee, Krithika Jagannath. Intelligent career advisors in your pocket? Americas Conference on Information Systems, 2019.

[18] Jeet Upadhyaya Prof. Debarati Ghosal Ganesh Dongre, Farooque Vadraiwala. Web-based career counseling system with a chatbot. International Research Journal of Modernization in Engineering Technology and Science, 2021.

[19] Gurav, Uma, and Nandini Sidnal. "Predict stock market behavior: role of machine learning algorithms." Intelligent Computing and Information and Communication: Proceedings of 2nd International Conference, ICICC 2017. Springer Singapore, 2018.

[20] Gurav, Uma Prashant, and S. Kotrappa. "Sentiment aware stock price forecasting using an SA-RNN-LBL learning model." Engineering, Technology Applied Science Research 10.5 (2020): 6356-6361.

[21] Gurav, Uma, and DS KOTRAPPA. "Predict stock market's fluctuating behavior: Role of investor's sentiments on stock market performance." SSRG International Journal of Engineering Trends and Technology 68 (2020): 72-80.

[22] Rajendra Hole, Shreyas, and Agam Das Goswami. "Design GA PSO-Based High-Efficiency SEPIC DC-DC Converter for Context-Aware Duty Cycle Control." Electric Power Components and Systems (2023): 1-20.

[23] Goswami, A. D., Hole, S. R. (2024). Analysis and Comparison of the DC-DC converter with Soft Computing algorithm. EAI Endorsed Transactions on Scalable Information Systems, 11(2).

[24] Sandeep, S. R., Ahamad, S., Saxena, D., Srivastava, K., Jaiswal, S., Bora, A. (2022). To understand the relationship between Machine learning and Artificial intelligence in large and diversified business organizations. Materials Today: Proceedings, 56, 2082-2086.

Authorized licensed use limited to: University of New South Wales. Downloaded on November 01,2025 at 03:50:46 UTC from IEEE Xplore. Restrictions apply.